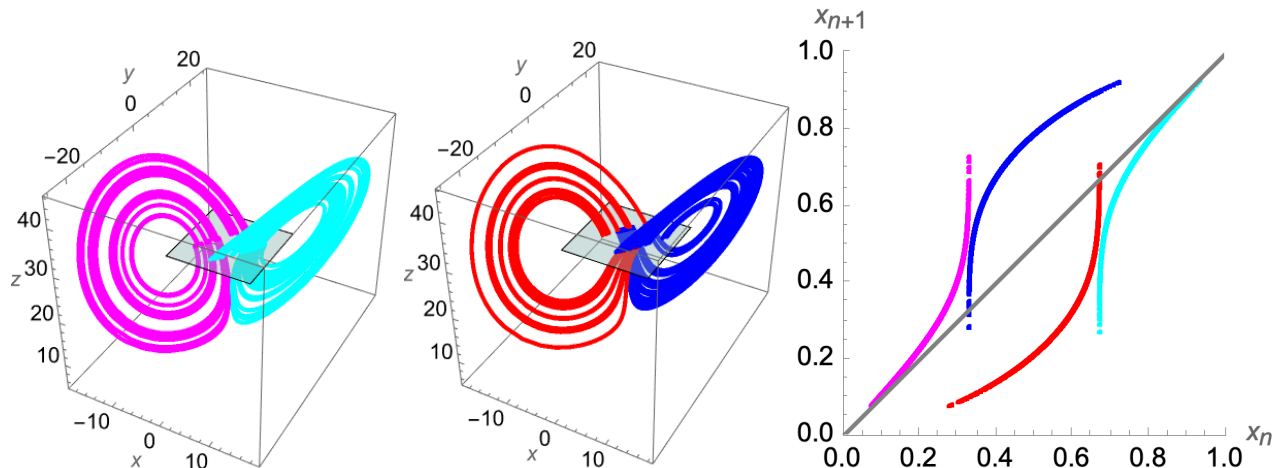


## Class 26 analyzing 1d maps

In the images below I split a **single** trajectory into pieces that start in the  $x < 0$  lobe and will traverse it again (magenta), that start in the  $x > 0$  lobe and will traverse it again (cyan), that start in the  $x < 0$  lobe and will switch (blue) and that start in the  $x > 0$  lobe and will switch (red).

Return map to  $z = 27$

(between the fixed points)



**Teams 7 and 8:** Post screenshots of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

### Questions

1. (Volume contraction) The Lorenz system is dissipative, meaning that volumes in the phase space are contracted under the flow. Consider an arbitrary closed surface  $S$ . This surface encloses a region  $W$  that has volume  $V$ . We can think of every point in  $W$  as the initial condition of a trajectory. Let each of them evolve forward in time (under the action of the dynamical system), let  $W(t)$  be the set they evolve to at time  $t$  (with surface  $S(t)$ ). The volume of the set is evolving in time!

The divergence of a vector field is a measure of local contraction (negative sign) or local expansion (positive sign) under the action of the vector field.

It turns out that  $\frac{dV}{dt} = \int_W \text{div } \underline{f} dV$

where  $\underline{f}$  is the vector field given by the dynamical system.

- (a) Find  $\text{div } \underline{f}$  and argue that  $\dot{V}$  is negative for the Lorenz system. Use this to conclude that volumes contract. When volumes in phase space are contracted under the action of the flow, we call a system *dissipative*, so you are showing that the Lorenz system is a dissipative system.

Recall that  $\text{div } \underline{f} = \nabla \cdot \underline{f} = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z}$ .

- (b) Use  $\dot{V} = \int_W \nabla \cdot \underline{f} dV$  to find  $V(t)$ , the evolution of  $V$ , for this system. What does  $V$  approach as  $t \rightarrow \infty$ ?

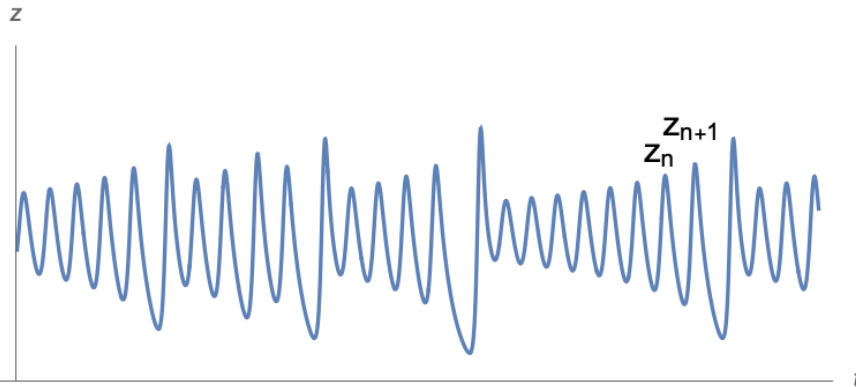
2. Show that  $f(x) = 3x(1 - x)$  on  $[0, 1]$  is conjugate to  $g(x) = x^2 - 3/4$  on a certain interval in  $\mathbb{R}$ , and determine the interval.

Because  $f$  and  $g$  are both quadratic, assume there is a linear change of coordinates  $h(x) = ax + b$ . Find  $a$  and  $b$  so that

$$h(f(x)) = g(h(x)),$$

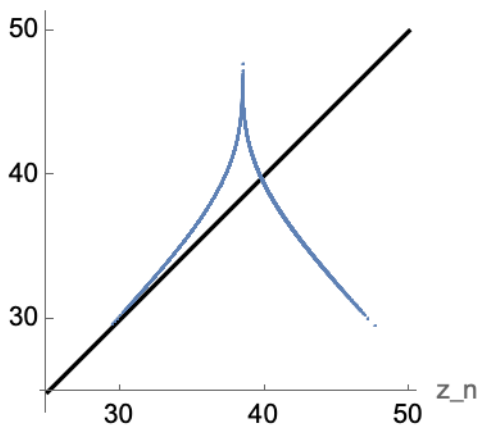
and then determine the corresponding interval for  $g$ .

*$f$  and  $g$  are both quadratic in  $x$ : choosing  $h(x)$  linear in  $x$  will work for this problem.*



The 'map', for a trajectory integrated to time 5000:

$z_{n+1}$



3. (9.4.2) The tent map is a simple analytical model that has some properties in common with the Lorenz map. Let

$$x_{n+1} = \begin{cases} 2x_n, & 0 \leq x_n \leq \frac{1}{2} \\ 2 - 2x_n, & \frac{1}{2} \leq x_n \leq 1. \end{cases}$$

- Sketch the graph of  $f(f(x))$ . How many times does it intersect the curve  $y = x$ ?
- Show the map has a period-2 orbit. This means that there is an  $x$  such that  $f(f(x)) = x$  (but  $f(x) \neq x$ ).
- Let  $g(x) = f(f(x))$ . Period-2 points are fixed points of  $g$ . Apply the derivative condition to  $g(x)$  and use the chain rule to classify the stability of any period-2 orbits.  
*Period-1 points are fixed points of  $g$  as well - why?*
- Look for a period-3 or period-4 point. If you find one, are such orbits stable or unstable?
- If you want, you can think about whether there is a period- $k$  orbit...